level 2

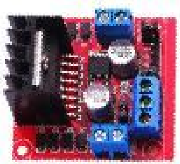
no. of sessions: 4

Energy Generating Wind Turbine

The wind turbine uses the power of the wind to generate electricity and provide clean energy. It is used in wind farms to supply electricity to homes

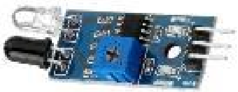
simple machine: gears
electronics: polarity

component gallery



Motor Driver

The brain of a robot. I listen to input devices like sensors and tell output devices like motors/LEDs when to turn ON/OFF.

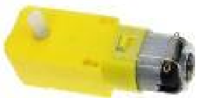


IR Sensor

I am like a robot's eyes. I help robots see objects or follow a path without touching anything.

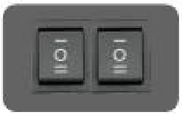
LDR Sensor

I am like a light detector who tells a robot if it is bright or dark and i am used in automatic lights.



Motor

I change electricity into movement and help wheels, fans, and robots spin and move.



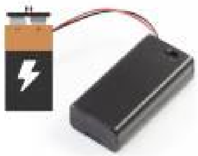
DPDT Switch Controller

Like a direction button - i make a motor spin forward or backward without using coding.

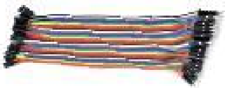
component gallery

Servo Motor
I am a **smart motor** that can stop at the exact position you want. I am used in robot arms, gates, and moving parts.

Servo Controller
I am like a **control room** for the servo motor and lets you move the servo easily to test how it works.



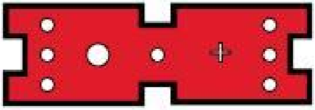
Battery & Battery Holder
I **give power** to the project, and my battery holder keeps the battery safe and connected. Together, we help the model work.

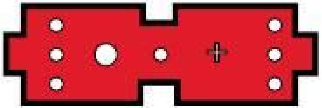


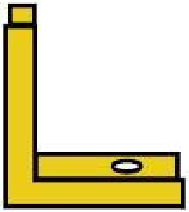
Jumper Wires
We are the **roads for electricity** who connect the different parts of the circuit, so power can travel.

Motor Clamp
I am like a **strong hand** that holds the motor tightly in place while it spins.

materials needed


F-F beam - x4

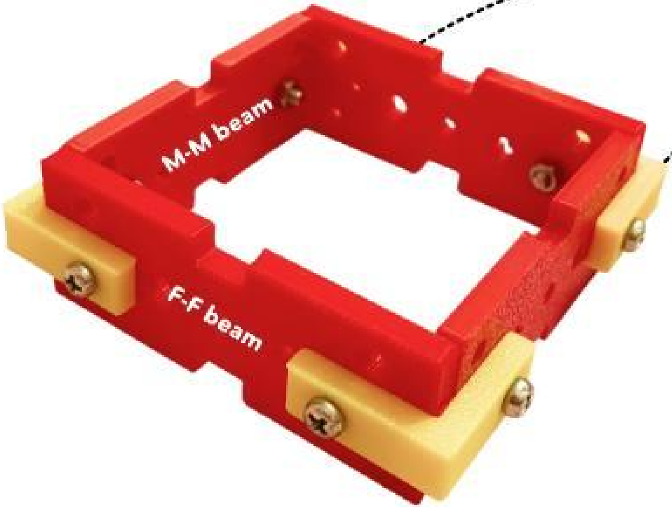

M-M beam - x4


L clamp - x8


small screw - x16


nut - x16

step 1 : assemble the base

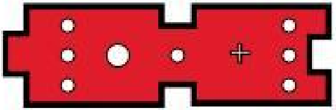


Use **4 Beams** to form a square shape.

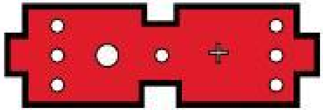
Use **L clamps** to hold them together.

🔁 **Repeat step 1 and make another square.**

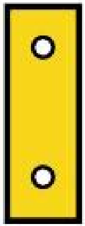
materials needed



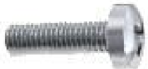
M- F beam - x2



M-M beam - x2



I clamp - x2



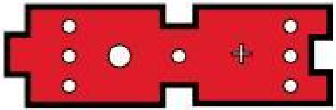
small screw - x4

nut - x4

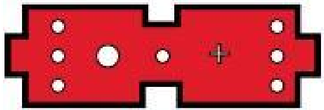
step 2 : add the supports

Use M-F beams and M-M beams joined together using I clamps.

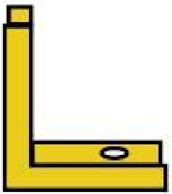
materials needed



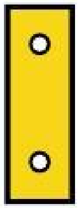
M-F beam - x3



M-M beam - x1



L clamp - x2



I clamp - x5

small screw - x12

nut - x12

step 3 : assemble the LED holder

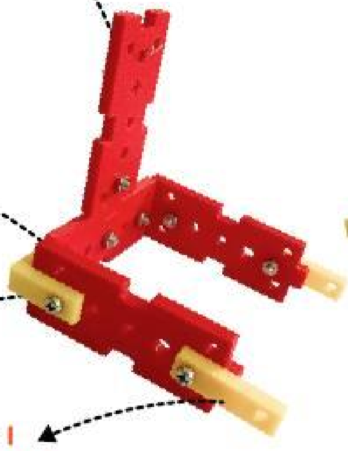
1.

Use **M-F beams** and **M-M beam** to form a C shape.

Attach another **M-F beam** vertically as shown

Use **L clamps** here

Use **I clamps** here



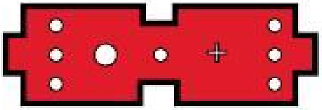
2.

Attach the **LED holder** to the base as shown

3.

Add to **I clamps** as shown

materials needed



M-M beam - x1

motor clamp x1

small screw - x6

nut - x6

step 4 : assemble the motor holder & clamp

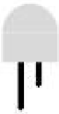
1. Fix the **second square** to the base as shown

2. Attach a **M-M beam** to the square as shown

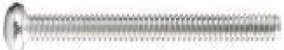
3. Fix a **Motor clamp** with a small screw

materials needed

motor x1



LEDs - x1



long screw - x2

nut - x2

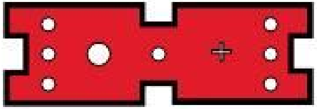
step 5 : add the motor clamp

Fix the **motor and LED.**

Fix the motor to the clamp using long screws as shown.

Pass the LED through hole on the beam.

materials needed



F-F beam - x4

small screw - x4

long screw - x4

nut - x8

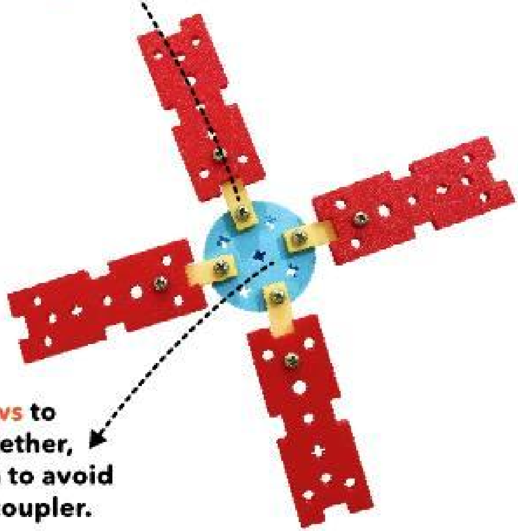
I clamp - x4

coupler - x1

step 6 : assemble the blades

1.

Use four F-F beams and fix it to the motor coupler using I clamps



2.

Use long screws to hold them together, Gently tighten to avoid breaking the coupler.

3.

Fix the blades to the motors axle.

step 7 : circuit assembly and testing

Fix the motor wire to the terminals of the LED.

Turn the blades really fast to make the LED glow.